

Lecture Notes V

Course: Functional Analysis

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Topic: Abstract integration

Theorem (Approximation Lemma). Suppose f is measurable and nonnegative, then define

$$A_{n,i} = \left\{ x \in X : \frac{i-1}{2^n} \leq f(x) < \frac{i}{2^n} \right\}, \quad B_n = \{x \in X : f(x) \geq n\},$$

$$\varphi_n(x) = \sum_{i=1}^{n2^n-1} \frac{i-1}{2^n} \chi_{A_{n,i}}(x) + n \chi_{B_n}(x).$$

It is clear that $\varphi_n(x)$ is simple for each $n \in \mathbb{N}$ and then for any fixed $x \in X$ we consider $\varphi_{n+1}(x) - \varphi_n(x)$ so that if $f(x) \in [(j-1)/2^n, j/2^n)$ we have $\varphi_n(x) = (j-1)/n$, as for $\varphi_{n+1}(x)$ notice that the interval $[(j-1)/2^n, j/2^n)$ is subdivided into two new intervals

$$[2(j-1)/2^{n+1}, 2j-1/2^{n+1}) \quad \text{and} \quad [2j-1/2^{n+1}, 2j/2^{n+1}),$$

if $f(x)$ belongs to the former [respectively, the latter], then

$$\varphi_{n+1}(x) = \frac{2(j-1)}{2^{n+1}} \quad \left[\text{respectively, } \varphi_{n+1}(x) = \frac{2j-1}{2^{n+1}} \right],$$

so that

$$\varphi_{n+1}(x) - \varphi_n(x) = \frac{2(j-1)}{2^{n+1}} - \frac{j-1}{2^n} = 0 \quad \left[\text{respectively, } \varphi_{n+1}(x) - \varphi_n(x) = \frac{2j-1}{2^{n+1}} - \frac{j-1}{2^n} = \frac{1}{2^{n+1}} \geq 0 \right].$$

We therefore conclude that (φ_n) is increasing. Given $\varepsilon > 0$, since for any $f(x)$ we can take $N_x > f(x)$ so that for $n > \max(N_x, \log_2 1/\varepsilon)$, $f(x)$ lies in some interval $[(k-1)/2^n, k/2^n)$, we thus have

$$|\varphi_n(x) - f(x)| = \left| \frac{k-1}{2^n} - f(x) \right| = f(x) - \frac{k-1}{2^n} \leq \frac{k}{2^n} - \frac{k-1}{2^n} = \frac{1}{2^n} < \varepsilon.$$

If $f(x)$ is bounded, say $|f(x)| \leq M$, then take $N_x = M + 1$ so that the convergence is uniform.

Theorem. If $\varphi, \psi : X \rightarrow \mathbb{R}$ are simple and nonnegative we have that

$$\varphi + \psi = \sum_{i,j} (\alpha_i + \beta_j) \chi_{A_i \cap B_j}$$

is the canonical representation of $\varphi + \psi$ so that

$$\int d\mu (\varphi + \psi) = \sum_{i,j} (\alpha_i + \beta_j) \mu(A_i \cap B_j) = \sum_{i,j} \alpha_i \mu(A_i \cap B_j) + \sum_{i,j} \beta_j \mu(A_i \cap B_j)$$

but since the B_j 's and the A_i 's are disjoint and form a partition for X , we have that

$$\mu(A_i) = \mu \left(\bigcup_j (A_i \cap B_j) \right) = \sum_j \mu(A_i \cap B_j)$$

and similarly for $\mu(B_j)$, yielding

$$\int d\mu (\varphi + \psi) = \sum_i \alpha_i \mu(A_i) + \sum_j \beta_j \mu(B_j) = \int d\mu \varphi + \int d\mu \psi.$$

Moreover, it is clear that

$$\int d\mu(a\varphi) = a \int d\mu \varphi$$

for any $a > 0$. If f, g are nonnegative and measurable, we have that there exist increasing sequences $(\varphi_n) \nearrow f$ and $(\psi_n) \nearrow g$ so that by the monotone convergence theorem

$$\begin{aligned} \lim \int d\mu(a\varphi_n + b\psi_n) &= \lim \left(a \int d\mu \varphi_n + b \int d\mu \psi_n \right) \\ &= a \lim \int d\mu \varphi_n + b \lim \int d\mu \psi_n = a \int d\mu f + b \int d\mu g. \end{aligned}$$

Theorem. Let $f : X \rightarrow \mathbb{R}$ be nonnegative and measurable, then if $f = 0$ μ -a.e., we have that

$$\mu(F = \{x \in X : f(x) > 0\}) = 0,$$

but notice how, given any h simple and nonnegative for which $h \leq f$ is satisfied, then $h > 0$ implies $f > 0$ so $H = \{x \in X : h(x) > 0\} \subseteq F$

$$\mu(H) \leq \mu(F) = 0 \implies \mu(H) = 0.$$

For $h = \gamma_j \chi_{\Gamma_j}$ and $\Phi_f = \{h : X \rightarrow \mathbb{R} \mid h \in \mathcal{E}^+, h \leq f\}$ we have

$$\int d\mu f = \sup_{\Phi_f} \int d\mu h$$

but for any $h \in \Phi_f$ we have

$$\int d\mu h = \sum_j \gamma_j \mu(\Gamma_j) = \sum_{j \neq z} \gamma_j \mu(\Gamma_j)$$

where z is the natural number for which $\gamma_z = 0$ (in case of existing) and thus for $j \neq z$ we have that $\Gamma_j \equiv h^{-1}(\gamma_j) \subseteq H$ and thus $\mu(\Gamma_j) = 0$ for all $j \neq z$ and then

$$\int d\mu h = 0$$

for any $h \in \Phi_f$, so we conclude that

$$\int d\mu f = 0.$$

Conversely, if $\int d\mu f = 0$ and $f \geq 0$, then let

$$B_n := f^{-1} \left(\left[\frac{1}{n}, \infty \right) \right)$$

so that

$$f(x) \geq \frac{1}{n} \chi_{B_n}(x) \implies \int d\mu f \geq \int d\mu \frac{1}{n} \chi_{B_n} = \frac{1}{n} \mu(B_n)$$

and hence

$$\mu(B_n) \leq n \int d\mu f = 0 \implies \mu(B_n) = 0 \implies \mu \left(\bigcup_{n \in \mathbb{N}} B_n \right) = 0$$

for all n , and finally, notice that $\{x \in X : f(x) > 0\} = \bigcup_{n \in \mathbb{N}} B_n$. We therefore conclude that $f = 0$ μ -a.e.

Theorem (Monotone Convergence Theorem). Suppose $(f_n) \nearrow f$ pointwise and f_n is measurable for all $n \in \mathbb{N}$, then since $f = \lim f_n = \sup f_n$, we have that f is measurable and $f_n \leq f$ for all n so that

$$\int d\mu f_n \leq \int d\mu f_{n+1} \leq \int d\mu f$$

so that the sequence

$$\left(\int d\mu f_n \right)$$

of integrals is increasing and bounded, so it is convergent and its limit is the supremum, we conclude that

$$\lim \int d\mu f_n = \sup \int d\mu f_n \leq \int d\mu f$$

because the integral of f is an upper bound for the sequence (f_n) .

Theorem (Fatou's Lemma). Suppose (f_n) is a sequence of measurable functions, then define

$$\phi_m = \sup F_m$$

with F_m the m -tail of the sequence $F = (f_n)$, and then ϕ_n is measurable and nonnegative for all $n \in \mathbb{N}$, but moreover, it is increasing by definition and by definition of $\liminf$ we have

$$\lim \phi_m = \liminf f_n.$$

The MCT ensures that

$$\int d\mu \liminf f_n = \lim \int d\mu \phi_m,$$

but the definition of ϕ_m implies that $\phi_m \leq f_k$ for $k > m$ so that

$$\int d\mu \phi_m \leq \int d\mu f_k$$

for $k > m$ so that

$$\int d\mu \phi_m \leq \ell_m$$

with $\ell_m = \inf \mathcal{F}_m$ and $\mathcal{F}_m$ the m -tail of $\mathcal{F} = (\int d\mu f_n)$. We therefore conclude that

$$\int d\mu \liminf f_n = \lim \int d\mu \phi_m \leq \lim \ell_m \equiv \liminf \int d\mu f_n.$$